



Abhijit Dalal

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B.Tech Electronics Engineering

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EDUCATION

•Veermata Jijabai Technological Institute (VJTI)

2027

B.Tech in Electronics Engineering

CGPA: 6.67

•Government Polytechnic Nashik

2024

Diploma in Electronics & Tele-communication Engineering

Percentage: 90.88

PERSONAL PROJECTS

•Skrolla | AI-Native Book Discovery App

Jun 2026 – Present

TikTok-style scrolling discovery for books: recommendations driven by embeddings and math, generative AI only where it belongs.

- Tools & technologies used: Kotlin, Jetpack Compose (MVVM), Python, CLIP, Sentence-Transformers, Supabase (pgvector), Cloudflare R2, Gemini via Firebase AI Logic
- Embedded ~91K books as 1152-dim vectors (CLIP ViT-L/14 on cover art + MiniLM on text, weighted concat); recommendations served by nearest-neighbor search over a per-user taste vector updated in real time from scroll, dwell, and save signals.
- Shipped the native Android app end-to-end, the Supabase + pgvector backend, and “StoryMode,” an LLM chat layer that lets users ask questions about a book before committing to it.

•MCP-Audit | Static Security Scanner for MCP Server Configs

Jun 2026

Scans Claude Desktop / Cursor MCP configs for security risks, every finding mapped to the OWASP MCP Top 10.

- Tools & technologies used: Python, FastAPI, Go, Next.js, OSV.dev API
- Built a static analyzer with 54 checks across 11 modules (secrets exposure, supply-chain attacks, prompt/tool poisoning, privilege escalation), covering all 10 OWASP MCP Top 10 categories, verified by 335 automated tests.
- Rebuilt the full engine natively in Go as an offline-first CLI validated to parity with the Python suite; exports SARIF 2.1.0 and CycloneDX 1.6 AI-BOM with live CVE lookups via OSV.dev. Deployed and live.

•Autonomous Racing using Deep RL | PPO Agents in 2D & 3D

Apr 2026 – May 2026

Reinforcement learning agents that learn to drive from scratch, first in a 2D simulator, then scaled into a full 3D game engine.

- Tools & technologies used: Python, PyTorch, Stable-Baselines3, Gymnasium, Unity ML-Agents, C#
- Trained a PPO agent (CNN policy, 4-frame stack) on Gymnasium CarRacing-v3 to consistently complete full laps from raw pixel input; scaled the pipeline into a 3D Unity ML-Agents environment with custom C# scripts for agent observations, reward shaping, and race-flow control.

•Harry Potter GPT | Full LLM Fine-Tuning Pipeline on nanoGPT

Feb 2026 – Mar 2026

A base GPT-2 taken through the complete modern LLM training recipe: pretrain, fine-tune, then align with human preference.

- Tools & technologies used: Python, PyTorch, HuggingFace Transformers, TRL, nanoGPT
- Engineered the 3-stage pipeline on GPT-2 (124M params): continued pretraining on ~5M tokens → supervised fine-tuning on 4,321 examples (val loss 2.33) → DPO (RLHF) alignment on 347 curated preference pairs; converted nanoGPT checkpoints to HuggingFace format, resolving Linear/Conv1D weight transposition for TRL compatibility.

EXPERIENCE

•Rishabh Instruments Limited

Jan 2024 – Apr 2024

Internship Trainee, Digital Production Department

Nashik

- Assisted in the assembly, testing, and calibration of digital metering instruments, ensuring compliance with quality standards.
- Diagnosed faults and troubleshooted electronic circuits on the production line, documenting processes throughout.

ACHIEVEMENTS

•Kaggle Bronze Medal, Google Play Store App Dataset 2026, 11K apps collected across 10 countries

2026

•Runner-up, Laghu Udyog Bharti (LUB), Hand Gesture Recognition System, real-time ASL recognition (OpenCV, MediaPipe, Raspberry Pi), diploma capstone

2024

POSITIONS OF RESPONSIBILITY

•Class Representative (CR), EXTC Engineering Diploma Class, Government Polytechnic Nashik

2021 – 2024